

Intelligence Beyond the Human

A Foundational Paper for SymK

EARLY REVIEW DRAFT | REVISION 2

Status: Early review draft, revision 2 - non-canonical and non-normative

Candidate edition: Foundation Paper v0.2

Review revision: 2.1.3H

Draft date: 20 August 2026

Historical source: SymK Foundation Paper Intelligence v0.1, preserved unchanged

Immediate predecessor: 2.1.3G Intelligence Foundation Paper v0.2 review draft, preserved unchanged

Decision dependency: Proposed SYMK-2X-DR-011; later reconciliation through SymK 2.1.4-2.1.9

Publication dependency: SymK 2.1.10 human review, cross-paper review, PDF generation, and publication acceptance

This document is the second interim human-readable synthesis requested by the project steward during SymK 2.1.3. It preserves the 2.1.3G Bearer correction and adds the multidimensional and specialization challenge. It is not the governed Foundation Paper v0.2 publication, does not accept DR-011 by implication, and does not create constitutional or concept-package authority. Its purpose is to expose the revised argument early enough for criticism and maturation.

Abstract

SymK needs a concept of Intelligence broad enough to include Human, artificial, collective, distributed, and hybrid cases without treating any one of them as the universal model. It also needs a concept precise enough to distinguish Intelligence from its substrate, particular skills, successful performances, stored representations, agency, authority, consciousness, and moral standing.

This paper proposes two deliberately separate statements.

The conceptual definition is:

Intelligence is a substrate-realized, multidimensional capacity of an organized system to integrate relevant resources in producing context-sensitive interpretations or effective responses across materially varying situations, including cases not individually specified in advance.

The specialization rule is:

An Intelligence specialization constrains the universal Intelligence concept to a declared Domain or family of situations and identifies the capability dimensions, relevant variation, evidence, limitations, and assessment conditions appropriate to that specialization.

The attribution rule is:

An Intelligence attribution identifies an organized system as the proposed Bearer, claims that the system actually bears Intelligence within a declared Scope, and records the relevant capability profile, Context, resources, dependencies, evidence, failures, alternatives, uncertainty, temporal validity, and revision conditions.

The first statement concerns the capacity itself. The specialization rule concerns a constrained form of that same concept, not a separate foundational Intelligence. The attribution statement concerns a fallible, governed claim that a particular system bears the capacity. Their separation is essential. A classification does not create a specialization; a profile does not create the multidimensional capacity it represents; a record does not create Intelligence; and one successful performance does not establish a capacity across variation.

The proposal preserves the non-anthropocentric and substrate-aware direction of the v0.1 paper while correcting five weaknesses. It makes multidimensionality constitutive rather than merely a measurement preference, removes Knowledge from the minimum definition, distinguishes the actual Bearer relation from a fallible attribution of a proposed Bearer, separates flexible response from online learning, and represents the Knowledge-Intelligence relationship through directional and contingent relations rather than circular definitions.

Contents

Abstract	1
1. Why Intelligence is foundational for SymK	4
2. The Copernican correction	4
3. The conceptual architecture	5
4. Substrate independence without substrate indifference	8
5. What the major research traditions contribute	9
6. The revised working definition	11
7. The attribution rule	13
8. Intelligence and adjacent concepts	15
9. Dimensions, specializations, profiles, and Projections	17
10. Intelligence and Knowledge	21
11. Individual, collective, distributed, and hybrid Intelligence	22
12. Intelligence in cooperation and Domain Intelligence Amplifiers	24
13. Semantic and engineering consequences	26
14. Candidate foundational commitments	28
15. Practical attribution gate	29
16. Counterexamples the definition must survive	30
17. Open questions and preserved dissent	33
18. Development requirements	34
19. Conclusion	35
20. Review status and path to publication	36
References	37
Document governance note	39

1. Why Intelligence is foundational for SymK

SymK exists to support Domain Intelligence Amplifiers: governed arrangements in which diverse participants work through representations of Domain Knowledge to produce better answers, faster and at lower cost, while preserving responsibility, provenance, challenge, and revision.

That identity cannot be engineered responsibly if Intelligence remains an unexplained compliment, a marketing label, a Human resemblance test, or a score. SymK must be able to ask:

- What capacity is being claimed?
- Which organized system is claimed to have it?
- Across which situations and variations?
- What resources and dependencies make the performance possible?
- What evidence supports the claim, and what failures constrain it?
- Who made and assessed the attribution?
- For which use is the attribution authorized?
- When must the attribution be reconsidered?

These questions are not secondary metadata. They prevent SymK from converting a useful output, a fluent interface, an institutional reputation, or an accepted record into an unqualified Intelligence attribution.

At the same time, SymK has bounded philosophical jurisdiction. It needs enough conceptual precision to govern its semantics and systems. It does not need to settle every metaphysical dispute about mind, subjective experience, life, or moral status.

2. The Copernican correction

2.1 Human Intelligence is an instance, not the center

Human Intelligence is historically the best-studied manifestation of Intelligence. Human language, embodiment, development, sociality, culture, emotion, and mortality provide indispensable evidence. They do not follow from that evidence as necessary conditions for every possible Intelligence.

A SymK definition must therefore avoid:

- treating adult Human cognition as the default architecture;
- defining Intelligence through biological origin;
- requiring Human-like language, affect, goals, or self-description;

- denying capability because its mechanism is unfamiliar;
- and inferring Human equivalence from functional similarity.

The correction is Copernican because it changes the reference point. Instead of asking how closely another system resembles a Human, SymK asks which organized capacity the system realizes, within which Context and relevant variation, and with what evidence.

2.2 Non-anthropocentric does not mean equivalent

Refusing Human privilege does not erase differences among Humans, other biological organisms, artificial systems, institutions, collectives, and hybrids. Systems may differ in:

- substrate and architecture;
- perception and action channels;
- learning and developmental history;
- temporal scale;
- energy and computation;
- memory and access to external resources;
- robustness, transfer, and failure behavior;
- explainability and self-monitoring;
- agency, autonomy, consciousness, and moral standing.

These differences belong in the capability profile and attribution evidence. They should neither disqualify a system by origin alone nor be hidden behind one undifferentiated label.

3. The conceptual architecture

The v0.1 paper proposed a useful layered view but placed attribution in the same sequence as causal realization. The revised model separates the actual phenomenon from claims and records about it.

3.1 Realization and manifestation

```
substrate
  ↓ realizes
organized system
  ↓ bears
multidimensional capacity configuration
  ↓ manifests under conditions
performance
```

3.2 Attribution and governance

```

observed and counterfactual evidence
  ↓ supports or challenges
Intelligence attribution claim
  ↓ is evaluated through
attribution assessment
  ↓ may authorize a bounded use and be preserved as
attribution record

```

The two sequences interact, but they are not identical. The organized system and its capacities can exist without a SymK record. Conversely, a record can exist while its claim is mistaken, outdated, exaggerated, or fraudulent.

3.3 Substrate

A substrate is the material or computational basis through which a capacity is realized. Biological tissue, electronic hardware, coordinated Human activity, institutional arrangements, communication networks, and combinations of these may participate in realization.

Substrate matters because it affects latency, memory, energy, durability, failure modes, modifiability, embodiment, and access to the environment. It does not by itself establish or deny Intelligence.

3.4 Organized system

The organized system is the causal subject at which the candidate capacity is realized. Organization matters because a substrate or collection of components does not automatically possess the capacities produced by their coordination.

The system boundary must be justified and held stable across the attribution being tested. Tools, documents, retrieval services, Human collaborators, models, sensors, institutions, and communication channels cannot move opportunistically inside or outside the boundary merely to preserve a favorable result.

A system may be:

- individual;
- collective;
- distributed;
- hybrid;
- nested within a larger system;
- or temporally persistent despite controlled component replacement.

These possibilities do not remove the burden of establishing Identity and boundary conditions.

3.5 Capacity and capability profile

Intelligence is a multidimensional capacity, not a performance event. The Bearer has an actual capability configuration whose dimensions may differ in strength, breadth, integration, reliability, development, and failure behavior.

A capability profile is a structured Representation or claim about that actual configuration. It may describe interpretation, inference, planning, problem solving, learning, coordination, transfer, error detection, creation, communication, or other Domain-relevant dimensions. The profile can be incomplete, mistaken, outdated, or tailored to one use. It does not create the configuration it describes.

No universal scalar is assumed. A scalar score may be a useful Projection for a declared decision, but it cannot replace the underlying profile, weighting rule, test conditions, and uncertainty.

3.6 Performance

Performance is what a system does under particular conditions. It is evidence for or against capability, not the identity of Intelligence.

The same performance may arise from:

- a durable flexible capacity;
- narrow competence;
- memorization or lookup;
- training-data or answer leakage;
- luck;
- hidden Human assistance;
- an external tool;
- environmental regularity;
- or a measurement defect.

Responsible attribution therefore requires repeated and varied evidence, relevant failures, resource disclosure, and alternative explanations.

3.7 Bearer

Bearer is the relational role occupied by that which actually bears a property, capacity, condition, or other attributable predicate. The organized system does not become a different kind of object when it bears Intelligence; it stands in a bearing relation to the capacity realized through its organization.

Actual bearing does not depend on recognition. A system may bear Intelligence before anyone identifies, describes, assesses, or records it. Conversely, an attribution may identify a system as Bearer even when the claimed bearing relation does not actually obtain.

This yields three distinct statements:

1. the universal concept of Intelligence can be defined without naming a particular Bearer;
2. every actual instance of Intelligence must be borne by an organized system;
3. an Intelligence attribution identifies a proposed Bearer and fallibly claims that the actual bearing relation obtains.

The first statement concerns conceptual universality, not a bearerless instance of Intelligence. Intelligence cannot exist as an actual, free-floating capacity detached from that which realizes and bears it.

This distinction is indispensable to SymK's origin as a cooperation protocol. Intelligence does not cooperate or contribute as an abstraction. Organized systems bearing relevant capacities participate in cooperation. Individual participants may each bear Intelligence, while their organized cooperation may or may not also constitute a new collective or hybrid Bearer. Interaction alone does not establish that additional system-level bearing relation.

SymK must distinguish:

1. the actual bearing relation;
2. a fallible attribution claim;
3. an assessment of that claim;
4. authority to rely on the assessment for a stated use;
5. and the record that represents the claim and assessment.

This distinction prevents database rows, labels, scores, approvals, or publications from being mistaken for the capacity they describe.

4. Substrate independence without substrate indifference

The conceptual balance remains:

Substrate-independent in definition; substrate-sensitive in realization and evidence.

Substrate independence means that biological origin, silicon implementation, institutional composition, or Human participation is not a definitional requirement.

Substrate sensitivity means that an attribution must still explain how the capacity is realized and constrained. The same abstract capability may have materially different:

- speed and cost;
- memory and persistence;
- environmental access;
- error patterns;
- opportunities for learning;
- control surfaces;
- safety implications;
- and conditions of failure.

Functional equivalence in one Scope does not establish mechanism equivalence, universal transfer, consciousness, moral equivalence, or interchangeable governance.

5. What the major research traditions contribute

No single research tradition supplies a complete SymK definition. Each illuminates a different Projection.

5.1 Psychometric traditions

Spearman, Thurstone, Cattell, Carroll, Wechsler, Gardner, Sternberg, and related traditions investigate the structure of Human cognitive differences. The history cannot be described accurately as a simple replacement of one-dimensional Intelligence by unrelated types. Spearman's general factor, Thurstone's primary abilities, Cattell's fluid and crystallized abilities, and Carroll's hierarchical strata motivate different accounts of common variance, broad abilities, and narrow abilities.

Empirical work supporting hierarchical models and tests of Gardner-style multiple intelligences suggest that general covariance and domain-sensitive structure can coexist. Visser, Ashton, and Vernon found a large general factor across several proposed intelligences together with more modest non-general associations within many domains. SymK therefore should neither turn one Human statistical factor into the universal ontology of Intelligence nor assume that every named specialization is independent of all others.

Their limitation for SymK is not their value but their population and instrument dependence. A structure extracted from Human test performance cannot silently become the ontology of every Intelligence.

5.2 Developmental and learning traditions

Piaget, Vygotsky, skill-acquisition research, and later learning-efficiency approaches emphasize change through interaction, instruction, practice, and transfer. They show why developmental history and prior resources matter.

They do not prove that online learning is a universal constitutive condition. A system may realize a flexible capacity through a fixed architecture produced by biological evolution, cultural development, prior training, or engineering design.

5.3 Rational-agent and goal-achievement traditions

Decision theory and artificial-intelligence research often analyze effective action toward objectives. Legg and Hutter formalize breadth across environments, while Russell and Norvig organize a large engineering tradition around agents and rational action.

These approaches clarify performance, environment, objective, and generality. SymK nevertheless keeps Intelligence distinct from rationality, agency, autonomy, and one stable utility function. Interpretive, exploratory, collective, or contemplative capacities need not reduce to a single goal-maximization model.

5.4 Embodied, enactive, distributed, and extended approaches

Varela, Thompson, and Rosch emphasize embodied engagement. Hutchins shows cognition distributed across people and artifacts. Clark and Chalmers challenge simple internal boundaries. Woolley and collaborators provide evidence for group-level performance structure.

These traditions make system-boundary analysis unavoidable. They support neither the claim that every tool is part of the Intelligence nor the claim that Intelligence must stop at the skin or process boundary.

5.5 Behavioral and mechanism-sensitive cautions

Turing demonstrates the importance of operational comparison without requiring privileged access to internal states. Searle warns that behavioral success may underdetermine claims about understanding. Marr distinguishes levels of analysis, while Putnam and Fodor help motivate realization across different mechanisms.

SymK draws a practical lesson: behavior is indispensable evidence, but neither one behavior nor one explanatory level exhausts the attribution.

5.6 Learning efficiency and emerging formal definitions

Chollet emphasizes skill-acquisition efficiency and novelty relative to priors. Burkart and collaborators examine the evolution of general Intelligence. Ng proposes a recent formal contribution to the continuing definitional debate.

These approaches reinforce the need to disclose priors, resources, relevant variation, transfer distance, and the difference between accumulated skill and capacity. Their exact formalisms remain research inputs rather than automatic SymK commitments.

5.7 Emotional, social, and other specialized constructs

Emotional Intelligence, social Intelligence, and logical-mathematical Intelligence demonstrate why the vocabulary of types must be handled carefully.

The ability model developed by Mayer, Salovey, and Caruso treats emotional Intelligence through related capacities to perceive, use, understand, and manage emotional information. Other so-called mixed models combine abilities with personality, motivation, disposition, or preferred behavior. These are not automatically the same construct merely because they share a label.

Social Intelligence is likewise analyzed through multiple capacities involving social perception, interpretation, expression, communication, coordination, and navigation of social situations. It overlaps emotional processing but also concerns roles, expectations, institutions, norms, intentions, relationships, and conflict.

Logical-mathematical Intelligence may denote a broad specialization, a narrower reasoning dimension, or a family of competences depending on the claim and evidence. Correctly executing one formal procedure establishes neither a multidimensional specialization nor general Intelligence.

For SymK, these traditions provide candidate specializations and dimensions. Their names do not grant foundational status or prove construct validity.

6. The revised working definition

Intelligence is a substrate-realized, multidimensional capacity of an organized system to integrate relevant resources in producing context-sensitive interpretations or effective responses across materially varying situations, including cases not individually specified in advance.

This sentence is conceptual. It does not specify the record format, the assessing authority, or the evidence sufficient for a particular Domain.

6.1 Substrate-realized

The capacity is not disembodied. It is realized through some physical, biological, computational, social, institutional, or hybrid organization.

6.2 Capacity

The definition concerns a relatively stable possibility of performance under relevant conditions, not one observed output. Capacity can be strong, weak, narrow, broad, reliable, fragile, developing, or declining.

6.3 Organized system

The capacity belongs at the level whose organization causally supports it. A component may contribute without bearing the system-level capacity. A collection may aggregate outputs without constituting an integrated system.

6.4 Multidimensional

Multidimensionality belongs to the actual capacity, not merely to its assessment. A Bearer may integrate several distinguishable but interacting capabilities, with different strengths and limitations. No single observed dimension, task, or score is assumed to exhaust the capacity.

A dimension is an aspect or axis along which the capacity can vary. A specialization is a constrained form of Intelligence organized around a Domain or family of situations. A profile is a Representation of a particular Bearer's capability configuration. A score or class is a further Projection constructed for an assessment or decision.

Dimensions can correlate, share supporting mechanisms, or participate in a higher-order organization. Multidimensionality therefore does not require complete independence among dimensions and does not prohibit a well-supported general factor within an explicitly bounded population and measurement model.

6.5 Integrate relevant resources

Integration means that available resources participate in a coordinated production of interpretation or response. It is stronger than mere co-location or addition, but it does not require one architecture.

Relevant resources may include signals, internal structure, memory, learned dispositions, tools, representations, other participants, environmental regularities, and prior experience. The list is intentionally non-exhaustive. Knowledge and feedback may be important resources in many cases, but neither is inserted as a universal definitional ingredient.

6.6 Context-sensitive interpretations or effective responses

Intelligence can be interpretive, operative, or both. The disjunction avoids excluding systems whose relevant capacity is primarily to make sense of situations and systems whose relevant capacity is primarily to respond effectively.

Context sensitivity means that differences which matter to the situation can affect interpretation or response. It does not require perfect awareness of every Context dimension.

Effectiveness is never free-floating. It must be evaluated relative to a task, problem, constraint, or standard without equating effectiveness with legitimacy, morality, or authorization.

6.7 Materially varying situations

The relevant cases must differ in ways that could matter to success. Repetition of an identical case demonstrates repeatability, not necessarily Intelligence.

Variation is Domain- and capability-dependent. It may concern inputs, constraints, goals, participants, environment, representation, sequence, uncertainty, resource availability, or failure conditions.

6.8 Cases not individually specified in advance

The system must do more than replay a separately enumerated answer for every accepted case. This requirement protects the concept from collapsing into a lookup table or exhaustive script.

It does not require online learning, stochasticity, self-modification, or an architecture whose rules were not designed in advance. A deterministic and fixed system can qualify if its organization supports context-sensitive interpretation or effective response across materially varying cases that were not each individually encoded.

7. The attribution rule

An Intelligence attribution identifies an organized system as the proposed Bearer, claims that the system actually bears Intelligence within a declared Scope, and records the relevant capability profile, Context, resources, dependencies, evidence, failures, alternatives, uncertainty, temporal validity, and revision conditions.

This is a governance and representation rule, not a second definition of the capacity.

The qualifier proposed is epistemic, not ontological. The attribution does not create the Bearer or the bearing relation. It proposes which organized system occupies the Bearer role and exposes that claim to evidence, assessment, challenge, and revision.

7.1 Actual limits and declared Scope

Every capacity has actual limits whether or not they are known. An attribution states a declared Scope: the range in which the claimant says the predicate applies.

The declared Scope may be:

- accurate;
- too broad;
- too narrow;
- underspecified;
- obsolete;
- or contested.

SymK must preserve this difference. Recording a Scope does not cause the actual capacity to fit it.

7.2 Claim, assessment, authority, and record

An attribution claim says that an identified system bears a specified Intelligence profile. An assessment evaluates that claim using declared criteria and evidence. An authority may permit reliance on the assessment for a bounded purpose. An attribution record represents these acts and their lineage.

Acceptance for one purpose does not authorize every use. A model may be accepted for document triage and not for final legal judgment. A clinical collective may be trusted for one specialty and not another. Capability, epistemic standing, decision authority, operational control, responsibility, and affected-party standing remain distinct.

7.3 Evidence and counterevidence

An attribution should preserve:

- successful and failed performances;
- the variation tested;
- relevant priors and developmental history;
- tools, collaborators, data, time, computation, and permissions;
- interventions and component-removal tests when feasible;
- alternative explanations;

- uncertainty and confidence;
- and conditions that would restrict, revise, or supersede the claim.

Failure is not merely noise. It locates the boundary of the profile and may reveal that the chosen Bearer or Scope was wrong.

8. Intelligence and adjacent concepts

8.1 Capability

Capability is the broader category. Intelligence is a particular organized capacity characterized by integrated, context-sensitive interpretation or effective response across relevant variation. Not every capability is Intelligence.

8.2 Competence

Competence is reliable adequacy in a task class or practice. It may support an Intelligence attribution, and practical Knowledge may overlap with it. Narrow competence does not automatically establish a broader Intelligence profile.

8.3 Skill

A skill is an acquired or developed ability organized around an activity. Skills can contribute to Intelligence and can be evidence of it, but no single skill is identical to Intelligence.

8.4 Performance

Performance is an event or result under conditions. Intelligence is a capacity inferred fallibly through patterns of performance and other evidence.

8.5 Cognition

Cognition can name processes such as perception, memory, inference, attention, or representation. Intelligence concerns an organized capacity expressed through relevant variation. The categories may overlap without being identical.

8.6 Adaptation and learning

Situational adjustment, stateful adaptation, learning, development, cultural formation, biological evolution, and engineering design are different routes by which flexible capacity may arise or change.

Online learning is not a universal condition of Intelligence. Learning and adaptation remain important profile dimensions, developmental mechanisms, and sources of evidence.

8.7 Rationality

Rationality evaluates reasoning or action against a norm. It may be a capability dimension or assessment standard. Intelligence does not entail conformity to every norm of rationality.

8.8 Agency and autonomy

Agency concerns action attributable to an agent. Autonomy concerns independence from external direction or control. A system may contribute intelligent interpretation without agency, or act intelligently while depending heavily on tools, designers, institutions, or collaborators.

8.9 Understanding

Understanding may be a Knowledge mode, an Intelligence dimension, both, or a distinct epistemic achievement. The present definition does not settle the issue and does not infer understanding from fluent performance alone.

8.10 Consciousness and moral standing

No Intelligence attribution automatically establishes subjective experience, consciousness, sentience, rights, dignity, responsibility, or moral standing. Neither does the absence of evidence for these predicates prove their absence. They require separate inquiry and governance.

8.11 Wisdom and authority

Wisdom concerns sound judgment in relation to values, consequences, and lived or institutional understanding. Authority concerns legitimate standing to decide, approve, or act. Neither follows from capability alone.

9. Dimensions, specializations, profiles, and Projections

SymK should treat Intelligence as actually multidimensional and its attribution as profiled. This requires several levels that ordinary language often collapses into “types of Intelligence.”

Level	Meaning	Example
Intelligence	The universal foundational concept and minimum nucleus	The definition in Section 6
Capability dimension	A distinguishable aspect along which the actual capacity may vary	inference, interpretation, transfer, regulation
Intelligence specialization	The universal concept constrained to a Domain or family of situations	emotional, social, logical-mathematical Intelligence
Actual capability configuration	The multidimensional capacity borne by a particular organized system	a Bearer's uneven strengths, integration, and limits
Attributed capability profile	A fallible Representation of that actual configuration	a governed assessment of the Bearer
Score or classification	A purpose-specific Projection from evidence and profile dimensions	a test score, class, ranking, or benchmark result

9.1 Capability dimensions

A capability dimension identifies a meaningful aspect of Intelligence without asserting that the aspect is a separate Intelligence. Candidate dimensions may include:

- interpretive sensitivity;
- inference and problem solving;
- planning and anticipation;
- transfer across cases;
- learning efficiency;
- adaptation under change;
- error detection and correction;
- emotional perception and regulation;
- social interpretation and coordination;
- logical and mathematical reasoning;
- creation and exploration;
- communication and explanation;

- robustness near boundaries;
- resource efficiency;
- and calibrated recognition of limits.

This is an analytical inventory, not a canonical universal list. Some dimensions may decompose into narrower dimensions, overlap, correlate, depend on a Domain, or prove to be assessment artifacts.

9.2 Intelligence specialization

An Intelligence specialization constrains the universal Intelligence concept to a declared Domain or family of situations and identifies the capability dimensions, relevant variation, evidence, limitations, and assessment conditions appropriate to that specialization.

A specialization inherits the universal Intelligence nucleus. It does not become a separate foundational concept merely by receiving a familiar name.

A candidate specialization should establish:

1. the Domain or family of situations to which it applies;
2. the dimensions that distinguish its capability configuration;
3. the materially varying and non-individually-specified cases it must address;
4. the relevant Context, resources, and dependencies;
5. evidence that it is more than one skill, personality label, preferred behavior, or outcome;
6. its overlaps and differences relative to adjacent specializations and constructs;
7. its assessment, failure, uncertainty, and revision conditions;
8. and its inheritance from the universal Intelligence definition.

The word type should be avoided when it implies mutually exclusive boxes. Specializations may overlap, nest, correlate, and share mechanisms. A Bearer may bear several at once and to different degrees.

9.3 Actual configuration and attributed profile

The organized Bearer has an actual multidimensional capability configuration whether or not anyone has described it. An attributed capability profile is a structured claim or Representation about that configuration.

This repeats the actual-bearing distinction at the dimensional level:

actual multidimensional capacity configuration
 ≠ attributed capability profile
 ≠ score or classification

An assessment may omit a dimension, weight it badly, use an invalid instrument, or measure a transient performance. The profile and its Projections remain fallible.

Breadth and depth are independent. A system may be exceptionally capable in a narrow Domain, moderately flexible across many Domains, or broad but shallow. Generality should never be inferred from prestige, fluency, or one benchmark family.

9.4 General factors and hierarchical structure

Multidimensionality does not require that all dimensions be statistically or causally independent. A general factor may summarize covariance across performances in a defined population and assessment model. Broad and narrow abilities may also form a hierarchy.

SymK should treat such a factor as an evidence-bearing statistical or assessment Projection unless stronger ontological grounds are established. A Human psychometric general factor cannot silently become:

- the complete Identity of Intelligence;
- a universal scalar valid for every substrate;
- proof that specializations are unreal;
- or proof that a high score transfers to every Domain.

Conversely, naming several specializations does not prove their independence or invalidate evidence of common organization.

9.5 Emotional Intelligence

Emotional Intelligence is a candidate specialization whose relevant situations involve emotional phenomena. Under an ability interpretation, its profile may include:

- perceiving emotionally relevant signals;
- using emotional information in interpretation or response;
- understanding emotional patterns, transitions, and relations;
- and regulating or managing emotion-relevant processes.

The specialization must identify whether the Bearer is interpreting its own emotional state, another participant's state, public emotional signals, or some combination. It must also separate demonstrated abilities from personality, motivation, empathy, preferred style, moral concern, and social approval.

An artificial system could demonstrate an emotion-related interpretive or responsive capacity without that evidence establishing subjective feeling or consciousness. Conversely, experiencing emotion would not by itself establish emotional Intelligence across relevant variation.

9.6 Social Intelligence

Social Intelligence is a candidate specialization concerning materially varying social situations. Its dimensions may include:

- detecting and interpreting social cues;
- understanding roles, relations, intentions, expectations, and norms;
- communicating and coordinating with other participants;
- anticipating social consequences;
- navigating cooperation, disagreement, and conflict;
- and revising responses as social Context changes.

It overlaps emotional Intelligence but is not reducible to it. A Bearer may recognize emotion accurately while misunderstanding authority, institutions, irony, group dynamics, cultural conventions, or cooperative obligations.

Social effectiveness must not be equated with manipulation, conformity, popularity, morality, or legitimate authority. These require independent evaluation.

9.7 Logical and logical-mathematical Intelligence

Logical or logical-mathematical Intelligence may be modeled as a candidate specialization when the claimed capacity integrates abstraction, relation detection, inference, formal structure, quantity, proof, or problem solving across materially varying cases.

In a narrower claim, logical reasoning may instead be one dimension, competence, or skill. Executing a memorized algorithm, applying one proof pattern, or obtaining one correct answer does not establish the full specialization.

The label must therefore declare its level:

- dimension;
- competence or skill;
- specialized capability configuration;
- or Intelligence specialization.

9.8 Specialization ownership

If this model is admitted, SymK would govern the universal Intelligence concept, the distinction among dimension, specialization, actual configuration, attributed profile, and Projection, and the minimum governance requirements.

The relevant semantic authority would govern a specialization's substantive content. A Psychology or Human-capability package may govern an evidence-supported emotional-Intelligence construct. A derived Domain may define a legal, clinical, diagnostic, or engineering capability specialization. Derived definitions would have to inherit the universal nucleus, disclose their Domain restriction, and preserve lineage.

Widely used labels should enter SymK first as candidates. Popularity, educational usefulness, or commercial assessment does not substitute for concept admission and evidence.

10. Intelligence and Knowledge

10.1 Independent definitions

Intelligence and Knowledge must not define each other in a circle.

The earlier v0.1 definition included Knowledge among the resources integrated by Intelligence. That may describe many real systems, but it makes Knowledge appear universally constitutive and risks pairing:

- Knowledge is what Intelligence possesses or responsibly uses;
- with Intelligence is what creates or uses Knowledge.

The revised minimum definition therefore refers to relevant resources without requiring Knowledge as one of them.

10.2 Directional and contingent reciprocity

Independence does not mean isolation. The relationship is rich but must be expressed through typed, directional, scoped, and challengeable claims.

Intelligence may:

- enable the pursuit of Knowledge;
- contribute to epistemic achievement;
- interpret claims and representations;

- test and challenge grounds;
- connect previously separated material;
- apply Knowledge in a Context;
- express results through Representations;
- and revise claims, models, assessments, or records.

Knowledge may:

- enable or constrain intelligent activity;
- scaffold a capability;
- support learning and formation;
- specialize Intelligence in a Domain;
- and provide evidence about a Bearer's capability when appropriately used.

These relations are contingent. An Intelligence attribution does not automatically establish Knowledge. A Knowledge attribution does not automatically establish Intelligence.

10.3 Knowledge is not stored as a substance

Documents, databases, graphs, models, prompts, property assertions, and messages are Representations. They may express claims, preserve grounds, guide action, or support epistemic achievement. Their storage does not by itself store Knowledge as though it were a transferable material.

Similarly, an Intelligence profile, score, or attribution record is a Representation. Storing it does not store or create the capacity.

10.4 The practical-Knowledge overlap remains open

Practical Knowledge and Intelligence both invoke competence across relevant variation. Some cases may satisfy both predicates. The current analysis supports separate definitions and rejects automatic cross-attribution, but it does not claim to have eliminated every metaphysical overlap.

This dissent remains open for later 2.1 work and final concept-package design.

11. Individual, collective, distributed, and hybrid Intelligence

11.1 Individual cases

An individual Human, organism, or artificial system may be the Bearer when its own organization supports the claimed profile under the declared conditions.

11.2 Collective and distributed cases

A group or institution is not intelligent merely because its members are intelligent or because their outputs are aggregated. A system-level attribution requires evidence of organization such as:

- durable coordination;
- differentiated and interdependent roles;
- integrated memory or shared state;
- information routing;
- system-level feedback and error correction;
- persistence across individual contributions;
- and capabilities not adequately explained by one component alone.

Whether these conditions establish literal collective Intelligence or only an explanatory shorthand remains a legitimate dissent. SymK should preserve both the evidence and the level of claim.

11.3 Hybrid cases

A Human-AI arrangement may remain a sequence of tool uses, or it may become a more persistent hybrid system. Connection alone does not create a new Bearer.

Evidence for a hybrid Bearer may include:

- stable Identity through time;
- a justified boundary;
- reciprocal dependence or adaptation;
- integrated memory;
- coordinated objectives or task regulation;
- role differentiation;
- system-level monitoring;
- and counterfactual degradation when critical relations are removed.

Component replacement does not necessarily destroy the system, but persistence criteria must be declared rather than assumed.

12. Intelligence in cooperation and Domain Intelligence Amplifiers

The accepted SymK Product Vision distinguishes:

- **governed Domain Knowledge as the medium;**
- **cooperation as the process;**
- **Amplified Intelligence as the target capability;**
- **better answers, faster and at lower cost, under accountability, as the intended value.**

This corrects the v0.1 motto that called cooperation the objective. Cooperation can be valuable, but in SymK it is the governed process through which diverse contributions may produce responsible amplification.

12.1 Cooperation composes different profiles

SymK's origin as a cooperation protocol makes multidimensionality operationally central. Cooperation does not merely connect several quantities of a generic Intelligence. It may coordinate Bearers with different and complementary capability configurations.

Different participants may:

- propose;
- retrieve;
- interpret;
- test;
- challenge;
- connect;
- decide;
- authorize;
- act;
- experience consequences;
- and maintain the resulting system.

One participant may contribute logical analysis, another Domain experience, another social interpretation, another emotional sensitivity, and another retrieval or comparison capacity. No participant is assumed capable of every role, and capability does not decide authority.

Complementarity is not automatically beneficial. Profiles may conflict, duplicate one another, introduce incompatible assumptions, create communication loss, or leave a critical dimension absent. The

cooperation protocol must therefore route contributions, preserve provenance, expose disagreement, and support correction.

12.2 A cooperative profile is not a sum

The profile of a cooperative system cannot be obtained merely by adding participant scores. System-level capacity depends on organization:

```
participant capability configurations
  ↓ coordination, translation, challenge, and integration
cooperative system capability configuration
  ↓ manifests under relevant variation
cooperative performance
```

A cooperative arrangement may exhibit:

- complementarity, when distinct strengths become jointly available;
- compensation, when one participant detects or corrects another's failure;
- emergence, when organization enables a capacity unavailable to any participant alone;
- interference, when coordination degrades a participant's contribution;
- masking, when one strong dimension hides failure in another;
- or fragility, when apparent capability depends on one unrecorded component.

These are empirical and governed possibilities, not automatic consequences of connection.

12.3 Amplification is multidimensional change

Amplified Intelligence should be represented as a change in capability configuration, not as an unqualified scalar increase.

```
baseline profile
  ↓ governed intervention or cooperation
resulting profile
  = gains, losses, unchanged dimensions,
    new integration, new dependencies, and uncertainty
```

A system may improve analytical speed while losing contextual sensitivity, social judgment, robustness, or error awareness. It may gain one new specialization while becoming more dependent on an external participant. Whether this qualifies as amplification depends on the declared target profile, Scope, consequences, trade-offs, and governing values.

More Knowledge, more stored content, greater access, one better answer, or faster performance does not by itself establish Amplified Intelligence. A later SymK 2.1.7 analysis must identify the candidate

amplified Bearer and show an increased integrated capacity across relevant variation under stable boundaries and accountable conditions.

13. Semantic and engineering consequences

13.1 Preserve the semantic declaration and computational Projection

The conceptual definition and attribution rules belong to SymK's governed semantic lane. Schemas, classes, JSON documents, database tables, APIs, and SharePoint columns are computational Projections.

A Projection may enforce or expose part of the meaning. It is never the meaning itself and must not acquire semantic authority merely because software depends on it.

13.2 Do not use one conflating Intelligence object

The v0.1 candidate IntelligenceObject inventory was useful as a completeness checklist, but one container risks conflating:

- the organized system;
- the actual capacity;
- the capability profile;
- a performance;
- an attribution claim;
- an assessment;
- authority for use;
- and the record Representation.

Engineering packages should keep these roles distinguishable even when an implementation stores them together for convenience.

13.3 Minimum attribution preservation

A material Intelligence attribution should preserve, proportionate to consequence:

- predicate identity and version;
- Bearer identity and stable boundary;
- substrate and organization claims;

- capability profile;
- Domain, declared Scope, Context, and time;
- tested variation and novelty;
- priors, developmental history, resources, assistance, and dependencies;
- successes, failures, counterfactuals, and alternatives;
- claimant, assessor, evidence sources, and governing authority;
- status, intended use, uncertainty, challenge, revision, and lineage.

Domain-specific projects own operational dimensions, thresholds, tests, and projections. They may specialize the universal minimum but must not silently weaken it.

13.4 Represent specialization without a flat type list

An implementation should not reduce emotional, social, logical, or future specializations to a mutually exclusive IntelligenceType enumeration. The computational Projection should support:

- one universal Intelligence concept identity and version;
- specialization identity, version, semantic authority, and lineage;
- declared Domain or situation family;
- inherited universal conditions;
- specialization-specific dimensions and relations;
- overlap, nesting, and non-exclusion among specializations;
- Bearer-specific attributed profiles;
- evidence and assessment models;
- and purpose-specific scores or classifications as separate Projections.

The structured semantic declaration of a specialization belongs in the appropriate governed semantics package. Runtime assertions state that a proposed Bearer satisfies that specialization within a declared Scope; they do not redefine the specialization.

13.5 Avoid bare Boolean attribution

A field such as `intelligent = true` is insufficient for any consequential use. It hides the profile, Scope, basis, authority, uncertainty, time, and revision conditions that give the assertion meaning.

The same prohibition applies to `emotional_intelligence = true` or `intelligence_type = social`. Consequential uses require the specialization, dimensions, criteria, evidence, profile, Scope, and uncertainty to remain recoverable.

13.6 Separate attribution from deployment authorization

Even a well-supported capability claim does not authorize deployment, autonomous operation, access to data, final decision-making, or control over affected parties. Those acts require independent governance.

14. Candidate foundational commitments

These commitments summarize the current proposal. In this review draft they remain epistemic candidates, not Ratified axioms or final concept specifications.

I-01 - Substrate distinction Intelligence is not identical to its substrate.

I-02 - Realization Every actual Intelligence is borne by an organized system and realized through that system's organized substrate.

I-03 - Organized-system level The capacity is borne at the system level whose organization supports it. An attribution must identify that proposed Bearer and justify its boundary and Identity.

I-04 - Non-anthropocentric scope Human Intelligence is one family of manifestations, not the universal template.

I-05 - Origin neutrality Biological, artificial, social, or hybrid origin neither proves nor disproves Intelligence or privilege.

I-06 - Capacity-performance distinction Performance is evidence for capacity and is not identical to Intelligence.

I-07 - Multidimensionality, profile, and Scope Actual Intelligence is multidimensional. Intelligence attributions represent that capacity through profiles which are scoped, contextual, temporal, evidence-bearing, uncertain, challengeable, and revisable.

I-08 - Relevant variation Intelligence requires flexible interpretation or effective response across materially varying cases, including cases not individually specified in advance.

I-09 - Learning non-necessity Online learning is not universally constitutive; adaptation, learning, formation, evolution, and design are distinct ways capacities may arise or change.

I-10 - Knowledge independence and reciprocity Knowledge and Intelligence are independently defined and related through directional, typed, scoped, and contingent relations.

I-11 - No automatic adjacent entailments Intelligence does not automatically entail rationality, agency, autonomy, understanding, consciousness, wisdom, moral standing, authority, authorization, or responsibility.

I-12 - Collective and hybrid burden Collective, distributed, and hybrid attributions require system-level organization and evidence beyond connection or aggregation.

I-13 - Attribution-layer distinction Actual bearing, attribution of a proposed Bearer, assessment, authority for use, and attribution record are different. Actual bearing may exist without recognition; attribution may be mistaken.

I-14 - Representation non-identity No label, score, schema, record, stored artifact, or accepted publication creates Intelligence by representation or declaration.

I-15 - Specialization inheritance An Intelligence specialization constrains the universal concept to a Domain or family of situations and must inherit its minimum nucleus while declaring specialized dimensions, variation, evidence, limitations, and assessment conditions.

I-16 - Specialization non-exclusivity Intelligence specializations may overlap, nest, correlate, and share mechanisms. Their names do not establish mutual exclusion, independence, or separate foundational status.

I-17 - Scalar non-reduction A general factor, score, class, or ranking may be a useful evidence-bearing Projection for a declared purpose but is not identical to Intelligence or to a Bearer's multidimensional capability configuration.

15. Practical attribution gate

Before accepting an Intelligence attribution for a consequential use, reviewers should ask:

Gate	Question
Predicate	Which version of the Intelligence meaning and which profile dimensions are being used?
Specialization	Is a specialization claimed, which universal conditions does it inherit, and which Domain or situation family constrains it?
Dimensions	Which dimensions are claimed, how are they related, and which relevant dimensions were omitted?
Bearer	Exactly which organized system is claimed to bear the capacity?
Identity	What makes it the same system across the attribution period?
Boundary	Why are these components inside and others treated as resources or environment?
Realization	Which substrate and organization support the claimed capacity?
Profile	How does the attributed profile represent the actual capability configuration, and which capabilities are claimed or excluded?
Domain	Within which field of activity or inquiry is the claim made?

Gate	Question
Scope	Across which tasks, environments, conditions, and times is the claim declared to apply?
Context	Which surrounding circumstances materially affect interpretation of the claim or evidence?
Variation	Which materially varying and not individually specified cases were tested?
Priors	What inherited, pretrained, designed, cultural, or institutional structure was available?
Resources	Which tools, representations, collaborators, data, time, computation, and permissions supported performance?
Evidence	Which repeated successes, failures, interventions, and counterfactuals support or limit the claim?
Alternatives	Could lookup, memorization, leakage, luck, imitation, hidden assistance, external control, or mere aggregation explain the result?
Reliability	How stable is the capacity, and how does it behave near its limits?
Projection	Is a score, class, general factor, or type label being mistaken for the multidimensional capacity or specialization?
Adjacent predicates	Are Knowledge, agency, autonomy, rationality, understanding, consciousness, authority, responsibility, or moral standing being inferred without separate grounds?
Governance	Who claims, assesses, authorizes use, may challenge, and must revise the attribution?
Time	When does the assessment expire or require reconsideration?
Cooperation	Which epistemic or operational roles can this profile responsibly support, and how does it complement or interfere with other profiles?

Passing the gate does not reveal an essence or establish universal generality. It establishes that the attribution is explicit enough to be examined, used within a bounded purpose, challenged, and revised.

16. Counterexamples the definition must survive

16.1 Exhaustive lookup

A table returns the correct answer for every enumerated input. It may be useful and reliable, but without capacity across materially varying cases not individually specified, correctness alone does not establish Intelligence.

16.2 Fixed but flexible specialized system

A deterministic system does not learn online but interprets new cases within a Domain through organized rules and models. It can qualify if the cases are not individually encoded and the response remains context-sensitive across relevant variation.

16.3 Fluent one-off success

A model produces an impressive answer once. The output is evidence, but leakage, prompting, hidden retrieval, assistance, or luck may explain it. A capability attribution requires a profile and broader evidence.

16.4 Narrow expert

A system performs exceptionally in one specialized Domain and fails outside it. It may bear narrow Intelligence within an honest Scope. Narrowness is not absence, and expertise is not universal generality.

16.5 Coordinated aggregation

A group votes or concatenates outputs. Better aggregate performance alone does not establish a new collective Bearer. Integration, persistence, feedback, and system-level causal evidence are required.

16.6 Distributed institution

No member possesses all relevant memory or capability, but organizational roles, records, communication, and correction sustain a durable capacity. A system-level attribution remains possible if the organization and boundary explain the capacity better than member aggregation alone.

16.7 Tool-dependent Human

A Human performs intelligently using language, notes, software, and collaborators. Resource dependence does not negate Intelligence, but the attribution must disclose whether the claimed Bearer is the Human, a wider coupled system, or both at different levels.

16.8 Intelligent but unauthorized

A system has a well-supported capability profile but no legitimate standing to make the final decision. Intelligence does not supply authority.

16.9 Stored profile

A database contains a detailed Intelligence attribution. The record may be useful, false, or obsolete. Storage does not preserve the capacity itself and does not make the claim true.

16.10 More content without amplification

A cooperative arrangement gains access to more documents and produces one better answer. This may improve performance without establishing an amplified system-level Intelligence. The Bearer and increased capacity across variation still require evidence.

16.11 Uneven specialization profile

A Bearer reasons exceptionally across logical and mathematical problems but repeatedly misreads emotional and social Context. The strong logical specialization neither proves general superiority nor disappears because other dimensions are weak. The attribution must preserve the uneven configuration.

16.12 Emotional classification without emotional Intelligence

A classifier maps facial images to emotion labels accurately on one benchmark but fails under cultural, contextual, temporal, and interactional variation. The score establishes performance within tested conditions, not a complete emotional-Intelligence specialization.

16.13 General score with hidden dimensional failure

A cooperative system's aggregate score improves while its error detection, social sensitivity, or robustness declines. The scalar increase cannot by itself establish Amplified Intelligence because the multidimensional trade-off and target profile remain undisclosed.

16.14 Named type without independent construct

An assessment labels a Bearer socially intelligent because it is popular or emotionally intelligent because it reports empathy. The label may capture personality, status, conformity, learned scripts, or self-presentation rather than the claimed specialization. Construct boundaries and variation evidence remain necessary.

17. Open questions and preserved dissent

1. What exact minimum separates Intelligence from very broad but non-intelligent regulation or control?
2. How should material variation and non-individual specification be measured across Domains?
3. Can every actual Intelligence be said to possess at least practical Knowledge, even though attribution of one does not entail attribution of the other?
4. Is understanding a Knowledge mode, an Intelligence dimension, both, or a distinct epistemic achievement?
5. When is a collective or institution literally intelligent rather than usefully described as intelligent?
6. Which causal and persistence conditions justify a hybrid Human-AI Bearer?
7. How should profiles be compared without smuggling one Domain's priorities into another?
8. When does a change in components, model version, membership, or tools create a new system Identity?
9. How should Intelligence claims age as environments, resources, and capabilities change?
10. What additional governance follows if consciousness, sentience, or moral standing becomes a credible concern?
11. Which aspects of Amplified Intelligence belong to Intelligence itself and which belong to cooperation, responsibility, or institutional design?
12. Which parts of the attribution inventory are universal semantics and which belong only to later engineering standards?
13. Is there a substrate-independent higher-order organization shared by every Intelligence, or are observed general factors population- and instrument-bound?
14. Which dimensions belong to the universal semantic inventory, and which should exist only in specialized packages?
15. Under what evidence should emotional, social, or logical-mathematical Intelligence be admitted as an Intelligence specialization rather than a capability bundle, competence, or skill?
16. How should SymK represent overlap and covariance among specializations without collapsing them or pretending they are independent?
17. Which profile changes qualify as Amplified Intelligence when some dimensions improve and others decline?
18. How should cultural and institutional dependence enter the Context and evidence for emotional and social specializations?

These are governed uncertainties, not defects to hide. A final v0.2 paper must state which were resolved by later 2.1 work and which remain open.

18. Development requirements

Implementations derived from this foundation should:

1. distinguish substrate, organized system, actual bearing, multidimensional capacity configuration, attributed profile, performance, attribution claim, assessment, authority, and record;
2. identify the proposed Bearer, claim the bearing relation explicitly, and preserve a stable, justified boundary;
3. represent Intelligence as multidimensional and use attributed profiles before scalar scores;
4. distinguish capability dimensions, specializations, actual configurations, profiles, and score or class Projections;
5. support overlapping and nested specializations rather than a flat, mutually exclusive type enumeration;
6. preserve specialization identity, version, authority, Domain, inheritance, dimensions, criteria, evidence, and lineage;
7. make Domain, declared Scope, Context, relevant variation, and time first-class;
8. disclose priors, resources, assistance, and dependencies;
9. preserve successes, failures, alternatives, uncertainty, and counterfactual evidence;
10. support individual, collective, distributed, nested, and hybrid systems without assuming them;
11. distinguish situational response, adaptation, online learning, development, evolution, and design;
12. prevent anthropomorphic and substrate-origin shortcuts;
13. separate capability attribution from authority, authorization, responsibility, and moral standing;
14. preserve challenge, supersession, restriction, and lineage;
15. treat schemas and stored records as Projections and Representations rather than semantic authorities;
16. allow Domain-specific specialization without weakening the universal minimum;
17. test whether claimed amplification is a multidimensional and durable capacity change rather than more content, one improved performance, or one higher aggregate score;
18. and disclose gains, losses, unchanged dimensions, integration changes, dependencies, and uncertainty in any amplification claim.

19. Conclusion

The literature does not provide one definition that SymK can copy unchanged. Psychometrics reveals general covariance together with broad and narrow structure in Human capability differences. Multiple-intelligences, emotional-Intelligence, and social-Intelligence traditions expose Domain-sensitive dimensions and specializations while leaving construct boundaries and independence open to evidence. Developmental and learning traditions reveal how capacities change. Rational-agent and formal approaches clarify performance across environments. Embodied, extended, distributed, and collective approaches challenge simplistic system boundaries. Behavioral and mechanism-sensitive traditions show why evidence must be operational without being naive.

SymK can preserve these insights through a disciplined distinction.

Intelligence itself is:

a substrate-realized, multidimensional capacity of an organized system to integrate relevant resources in producing context-sensitive interpretations or effective responses across materially varying situations, including cases not individually specified in advance.

An Intelligence specialization is:

the universal Intelligence concept constrained to a declared Domain or family of situations, with identified capability dimensions, relevant variation, evidence, limitations, and assessment conditions appropriate to that specialization.

An Intelligence attribution is:

a governed and revisable claim that identifies an organized system as the proposed Bearer, claims that the system actually bears Intelligence within a declared Scope, and records the capability profile, Context, resources, dependencies, evidence, failures, alternatives, uncertainty, temporal validity, and revision conditions relevant to its use.

The first concerns what is claimed to exist. The specialization statement concerns a constrained form of the same concept. The attribution statement concerns how SymK responsibly makes, evaluates, records, authorizes, challenges, and revises a claim about a particular Bearer.

The resulting ontological constraint is categorical:

The universal concept of Intelligence does not require a particular Bearer to be named. Every actual Intelligence requires an organized system that bears it. Recognition and attribution of that Bearer are subsequent and fallible.

This distinction supports the larger SymK identity:

Governed Domain Knowledge is the medium. Cooperation is the process. Amplified Intelligence is the target capability. Better answers, faster and at lower cost under accountability, are the intended value.

Cooperation can combine complementary capability configurations, compensate for failures, and sometimes produce a new system-level capacity. It can also create interference, fragility, dependency, or loss in dimensions that an aggregate score conceals. Amplification must therefore be evaluated as a governed multidimensional change rather than a generic increase.

Human Intelligence remains a central source of evidence and an important participant in that process. It is not the measure of all possible Intelligence. Emotional, social, logical-mathematical, and future specializations may illuminate real dimensions without becoming mutually exclusive foundational Intelligences by naming alone. Artificial, collective, distributed, and hybrid cases remain possible, but none receives the attribution by origin, connection, fluency, storage, classification, score, or declaration alone.

20. Review status and path to publication

This second review draft incorporates the accepted wording recommendations, the Bearer challenge, the multidimensional-specialization challenge, and the analytical results available through SymK 2.1.3. It deliberately does not close the later dependencies assigned to:

- 2.1.4 - Knowledge engineering, epistemic conditions, and Representation;
- 2.1.5 - cooperation and cooperative formation;
- 2.1.6 - agency, authority, responsibility, and related predicates;
- 2.1.7 - Amplified Intelligence and the DIA bearer;
- 2.1.8 - principles and axiom consequences;
- 2.1.9 - integrated conceptual baseline;
- 2.1.10 - cross-paper revision, human review, PDF projection, and publication acceptance.

Before a governed v0.2 edition is published, reviewers must:

1. reconcile this draft with the accepted or revised disposition of DR-011;
2. confirm that the DR-010 Bearer vocabulary covers both actual bearing and fallible attribution, or enter a narrowly scoped clarification without silently reopening its wider structural decision;
3. decide whether multidimensionality belongs in the accepted Intelligence nucleus and determine the admission status of the specialization mechanism;
4. route emotional, social, logical-mathematical, and other candidate specializations to appropriate semantic authorities rather than accepting them by example;
5. apply the results of 2.1.4-2.1.9;
6. compare terminology across all four Foundation Papers;
7. verify every retained and added citation and literature characterization;
8. preserve v0.1 and 2.1.3G lineage and material dissent;
9. complete human-readability review;
10. generate and visually inspect the v0.2 PDF;
11. and record the 2.1.10 human approval checkpoint.

References

- Alloway, T. P., Argo, D., Gruskin, S., and Comer, B. (2026). "The Development of a Social Intelligence Test for Teenagers: A Pilot Study." *Children*, 13(4), 443. [Open article](#).
- Bickley, P. G., Keith, T. Z., and Wolfle, L. M. (1995). "The Three-Stratum Theory of Cognitive Abilities: Test of the Structure of Intelligence Across the Life Span." *Intelligence*, 20(3), 309-328. [Publisher record](#).
- Burkart, J. M., Schubiger, M. N., and van Schaik, C. P. (2017). "The Evolution of General Intelligence." *Behavioral and Brain Sciences*, 40, e195. [Publisher record](#).
- Carroll, J. B. (1993). *Human Cognitive Abilities: A Survey of Factor-Analytic Studies*. Cambridge University Press. [Publisher record](#).
- Cattell, R. B. (1943). "The Measurement of Adult Intelligence." *Psychological Bulletin*, 40(3), 153-193. [Publisher record](#).
- Chollet, F. (2019). "On the Measure of Intelligence." arXiv:1911.01547. [Open manuscript](#).
- Clark, A., and Chalmers, D. (1998). "The Extended Mind." *Analysis*, 58(1), 7-19. [DOI and manuscript](#).
- Fodor, J. A. (1974). "Special Sciences (Or: The Disunity of Science as a Working Hypothesis)." *Synthese*, 28, 97-115. [Publisher record](#).
- Gardner, H. (1983). *Frames of Mind: The Theory of Multiple Intelligences*. Basic Books.
- Gottfredson, L. S. (1997). "Mainstream Science on Intelligence: An Editorial with 52 Signatories, History, and Bibliography." *Intelligence*, 24(1), 13-23. [Publisher record](#).
- Hutchins, E. (1995). *Cognition in the Wild*. MIT Press. [Publisher record](#).

- Legg, S., and Hutter, M. (2007). "Universal Intelligence: A Definition of Machine Intelligence." *Minds and Machines*, 17, 391-444. [Open manuscript](#).
- Marr, D. (1982/2010). *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. MIT Press. [Publisher record](#).
- Mayer, J. D., Salovey, P., and Caruso, D. R. (2004). "Emotional Intelligence: Theory, Findings, and Implications." *Psychological Inquiry*, 15(3), 197-215. [Author repository record](#).
- Newell, A., and Simon, H. A. (1976). "Computer Science as Empirical Inquiry: Symbols and Search." *Communications of the ACM*, 19(3), 113-126. [Publisher record](#).
- Ng, K.-S. (2025). "On the Definition of Intelligence." arXiv:2507.22423. [Open manuscript](#).
- Piaget, J. (1952). *The Origins of Intelligence in Children*. International Universities Press. [Publisher record](#).
- Putnam, H. (1967). "Psychological Predicates." In W. H. Capitan and D. D. Merrill, eds., *Art, Mind, and Religion*. University of Pittsburgh Press. Reprinted as "The Nature of Mental States."
- Russell, S., and Norvig, P. (2020). *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson. [Book site](#).
- Salovey, P., and Mayer, J. D. (1990). "Emotional Intelligence." *Imagination, Cognition and Personality*, 9(3), 185-211. [Publisher record](#).
- Searle, J. R. (1980). "Minds, Brains, and Programs." *Behavioral and Brain Sciences*, 3(3), 417-424. [Publisher record](#).
- Spearman, C. (1904). "General Intelligence, Objectively Determined and Measured." *The American Journal of Psychology*, 15(2), 201-292. [Publisher record](#).
- Sternberg, R. J. (1985). *Beyond IQ: A Triarchic Theory of Human Intelligence*. Cambridge University Press.
- Thurstone, L. L. (1938). *Primary Mental Abilities*. University of Chicago Press.
- Turing, A. M. (1950). "Computing Machinery and Intelligence." *Mind*, 59(236), 433-460. [Original article](#).
- Varela, F. J., Thompson, E., and Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press. [Publisher record](#).
- Visser, B. A., Ashton, M. C., and Vernon, P. A. (2006). "Beyond g: Putting Multiple Intelligences Theory to the Test." *Intelligence*, 34(5), 487-502. [Publisher record](#).
- Vygotsky, L. S. (1978). *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press.
- Wechsler, D. (1944). *The Measurement of Adult Intelligence*, 3rd ed. Williams & Wilkins.
- Woolley, A. W., Chabris, C. F., Pentland, A., Hashmi, N., and Malone, T. W. (2010). "Evidence for a Collective Intelligence Factor in the Performance of Human Groups." *Science*, 330(6004), 686-688. [Publisher record](#).

Document governance note

Foundation Papers are epistemic sources and human communication artifacts. They do not become constitutional norms, final concept packages, formal ontologies, or implementation specifications through publication or location.

This draft preserves every v0.1 bibliography entry and broad line of inquiry while adding sources directly relevant to hierarchical ability structure, multiple-intelligences testing, emotional Intelligence, and social Intelligence. Source verification, cross-paper consistency, formal lineage mapping, human review, and PDF publication remain mandatory at 2.1.10. No implementation should silently promote the candidate specializations, profile dimensions, commitments, attribution gate, or engineering inventory in this draft into a canonical SymK schema.